CS 455 / CS 555 — Project Proposal

Large Language Models — Spring 2025/2026

Section 1 — Group Members

List all group members in the exact format below. Groups must have 2 or 3 members.

```
Mustafa_Ege_Özer, 32681
Umut_Can_Çubukçu, 32453
Yiğit Onur Yılmaz, 31981
```

Section 2 — Track

Our track: CS 455 Project + Paper Track Application

Our group is highly suited for the Paper Track due to our strong research backgrounds and domain expertise. Specifically, Umut Can Çubukçu brings extensive undergraduate research experience across 4 PURE projects, ranging from Camera Calibration for 3D Reconstruction to current work on Batch Optimization for Active Learning with Dr. Erhun Kundakcıoğlu. This technical foundation, combined with a genuine passion for sports analytics and prior experience handling complex football datasets (e.g., Transfermarkt), provides the ambition and practical skills necessary to elevate this Text-to-SQL scouting agent into a publishable academic contribution. Also, Yiğit Onur Yılmaz can contribute to this project with his Data Base Administration background where he acquired it from his ongoing long-term internship and he worked on a machine learning model which detects probable anomalies in database and informs the executives via mail. Mustafa Ege Özer contributes to our group with deep learning research experience, AI-heavy courseload, an ongoing paper publication process in sentimental image segmentation, a PURE project on 3D reconstruction with Gaussian Splatting and diffusion techniques supervised by Dr. Hüseyin Özkan, and author/reviewer roles at IEEE SIU 2026 Conference, providing our team with the necessary insight to design, develop and report a publishable research project.

Section 3 — Project Proposal

3.1 Project Title

ScoutRAG: A Retrieval-Augmented Football Player Scouting Assistant

3.2 Project Category

(A) Application / System Building

3.3 Problem Definition

Football scouts and technical staff often lack the database querying skills (e.g., SQL) required to perform complex, multi-variable filtering on massive tabular player databases. Our project tackles this gap by building an agentic Text-to-SQL framework that translates tactical scouting needs into executable database queries. The system takes complex natural language requests as input (e.g., "Find left-footed center-backs under 23 with passing > 15 and a market value under €10M"). It processes this against a 42,500+ row database and outputs a precise list of matching players alongside a brief analytical summary. Success for this project is defined as the system's ability to consistently generate accurate, executable SQL queries from natural language that return the exact tabular matches without fabricating any player attributes or ignoring user constraints.

3.4 Motivation

Football scouting requires comparing many players across technical, physical, mental, positional, and financial dimensions. Traditional search interfaces usually require users to manually set many filters and then interpret large tables of attributes. A natural-language scouting assistant can make this process more intuitive by allowing users to express football-specific needs in ordinary language and receive explainable recommendations. This is interesting for an LLM course because it combines retrieval, structured filtering, prompt engineering, and grounded generation. Users who could benefit include football fans, sports analytics students, amateur scouts, and game players who want interpretable player recommendations rather than only raw statistics.

3.5 Approach

We will build a hybrid Retrieval-Augmented Generation (RAG) pipeline tailored for tabular data. First, we will preprocess the Football Manager dataset by handling missing values and normalizing numerical fields (e.g., age, wage, attributes). We will then convert each player's row into a compact textual profile containing structured information and key attributes. To maximize retrieval accuracy, we will implement a dual-storage strategy: the exact structured data will be stored in a Pandas/SQLite database, while the textual profiles will be embedded using sentence-transformers and indexed in a vector database (FAISS or Chroma).

At query time, the system will utilize a hybrid retrieval strategy. Explicit constraints (e.g., age < 23, value < €10M) will be handled via structured SQL/Pandas filtering, while semantic and tactical requirements (e.g., "high-pressing midfielder" or "creative winger") will be resolved using vector retrieval. The top-k candidates from this intersection will be passed to an open-source instruction LLM (e.g., Llama-3 or Mistral via HuggingFace/Colab) through a strict prompt that grounds the generation entirely on the retrieved records. The final output will be a ranked scouting report featuring evidence-based attributes, tactical weaknesses, and alternative options. **Pipeline:** User query → constraint parsing → structured filtering + vector retrieval → top-k profiles → LLM reranking/explanation → grounded scouting report.

3.6 Data

We will utilize a custom-extracted tabular dataset sourced directly from the Football Manager 2024 (FM24) database. The dataset contains 42,541 player profiles across 57 structured numerical and categorical columns (e.g., age, position, value, wage, and 1-20 rated attributes) in English. The data was acquired strictly through the game's official, permitted in-game export functionality. To ensure a comprehensive dataset, in-game "attribute masking" was disabled, revealing the exact numerical ratings for every player. Since the game was legally obtained for free via Epic Games, we consider this high-quality dataset as highly accessible "mock data" to safely validate our academic prototype without violating any web scraping terms of service or intellectual property rights. For future iterations and potential commercial scaling beyond this course, we plan to substitute this mock data with professional, licensed football data providers (e.g., StatsBomb, Opta).

https://drive.google.com/file/d/1HDYY4EhaoJYxIeBtoSDlRuQWY6ZpqJ-U/view?usp=sharing

3.7 Evaluation Plan.

We will evaluate both retrieval quality and final answer quality. We will create a test set of approximately 30-50 scouting queries covering different roles and constraints (e.g., young striker, cheap goalkeeper, high-stamina fullback, creative playmaker). For each query, we will define expected constraints and, when possible, manually judge relevant candidates from the dataset. Retrieval will be measured using Recall@5, Recall@10, Mean Reciprocal Rank (MRR), and constraint- satisfaction rate. Generation will be evaluated with human ratings on relevance, factual grounding, usefulness of explanation, and hallucination rate. Baselines will include: (1) a structured-filter-only baseline, (2) a vector-retrieval-only baseline, and (3) an LLM-only baseline without retrieval. If time permits, we will run an ablation comparing structured filtering + vector retrieval versus vector retrieval alone.

3.8 Timeline

Week / Dates	Planned Milestones / Deliverables
Week 1 (now – May 10)	Finalize project scope, choose dataset, write proposal, create initial preprocessing notebook, inspect columns and data quality.
Week 2 (May 11 – May 17)	Clean dataset, create player text profiles, implement structured filtering, build first vector index with sentence-transformer embeddings.
Week 3 (May 18 – May 24)	Implement RAG pipeline, prompt templates, ranked scouting output, and initial Streamlit or notebook demo.
Week 4 (May 25 – May 31)	Create evaluation query set, run baselines, compute retrieval metrics, perform error analysis and ablation experiments.
Week 5 (June 1 – June 7)	Finalize experiments, polish demo, write final report,prepare README, clean code Repository, and prepare presentation/demo materials.

3.9 Risks and Contingency Plan

Data-related Risk: The selected Football Manager dataset may contain missing or inconsistent values for highly variable fields such as market value, wage, or specific positional familiarity.

- *Plan B:* We will implement a robust preprocessing step to clean and normalize the data. If financial fields prove too unreliable for SQL filtering, we will temporarily exclude them from the evaluation phase and focus the retrieval strictly on robust, fully populated fields (e.g., age, primary position, and 1-20 rated attributes).

Technical Risk (Hallucination): The LLM might hallucinate player facts, invent attributes, or overstate scouting evidence when generating the final analytical report.

- *Plan B:* We will strictly enforce grounded prompting by passing *only* the retrieved structured rows into the LLM's context window. Furthermore, we will implement an automated post-generation check to verify that the numbers cited in the text perfectly match the retrieved source records before presenting the output to the user.

Compute-related Risk: Running an open-source LLM for both SQL generation and final report synthesis might be excessively slow or exceed Google Colab's Free Tier GPU limits.

- *Plan B:* We will optimize the pipeline by restricting the top-k retrieval size, caching frequent vector outputs, and utilizing heavily quantized models (e.g., 4-bit). If local generative inference remains a bottleneck, we will offload only the final text-generation step to a fast, free-tier cloud API (e.g., Groq) while keeping all SQL and vector retrieval operations completely local.

3.10 Compute Resources

Which compute will you use? Be honest about your access — this helps validate that the scope is realistic.

- [X] Google Colab (free tier)
- [X] Kaggle Notebooks
- [ ] Free credits from Google / other platforms (specify)
- [ ] Paid APIs — OpenAI / Anthropic / other (cost is your own responsibility)
- [X] Local hardware (specify GPU)
- [ ] Other (specify)

The project scope is realistic for free compute because the main components are data preprocessing, embedding generation, vector indexing, retrieval, and prompt-based generation. Embedding 150k player profiles can be done in batches on Colab/Kaggle, and the vector index can be stored locally. We do not plan to fine-tune a large model as the core requirement; if generation with a local model is too slow, we will use a smaller open-source model or limited paid/free API calls for the final answer generation while keeping evaluation reproducible.

3.11 Expected Outcomes

What do you expect to deliver by June 7? What will the report contain? What will the demo show?

By June 7, we expect to deliver a fully functional, hybrid RAG scouting assistant, accessible via a Streamlit or notebook interface. The system will accept natural-language scouting queries and return ranked, explainable shortlists of players grounded strictly in retrieved data via structured filtering and semantic vector search. Our final academic report (~12-15 pages, fulfilling Paper Track expectations) will detail the dataset preprocessing, system architecture, retrieval metrics computed from our custom evaluation query set, ablation experiments, and a rigorous error analysis. The polished code repository will include all preprocessing notebooks, the dual-retrieval pipeline, evaluation scripts, and a clear README for reproducibility. Finally, our presentation and demo materials will showcase the system handling nuanced queries—such as “cheap young striker” or “defensive midfielder for high pressing”—displaying the exact retrieved evidence behind every recommendation.

Section 4 — References (optional but encouraged)

List any papers, datasets, or tools you plan to build on. Required for CS 555 and Paper Track applicants.

Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.
https://arxiv.org/abs/2005.11401
Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.
https://arxiv.org/abs/1908.10084
Johnson, J., Douze, M., and Jegou, H. (2017). Billion-scale similarity search with GPUs / FAISS.
https://arxiv.org/abs/1702.08734
Kaggle. Football Manager complete data (150k+).
https://www.kaggle.com/datasets/furkanuluta/football-manager-22-
complete-player-dataset
Kaggle. Football Manager Data (150000+ players).
https://www.kaggle.com/datasets/ajinkyablaze/football-manager-data
HuggingFace Sentence Transformers documentation. https://www.sbert.net/
FAISS documentation. https://faiss.ai/